

SHAHZEB MAHMOOD

DevSecOps & Platform Engineer

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SUMMARY

DevSecOps and Platform Engineer with an MSc in Computer Science with Cyber Security, specialized in high-compute AI workloads and infrastructure automation. Experienced in scaling high-performance computing (HPC) clusters, managing deployments of NVIDIA GPUs, and designing secure CI/CD pipelines. Proven track record shifting security left, improving delivery metrics, and optimizing performance for resource-intensive machine learning data pipelines.

SKILLS

- Infrastructure & Compute:** High-Performance Computing (HPC), NVIDIA GPU Workloads, AWS, Azure, GCP, Kubernetes, Docker, Terraform, Pulumi, Helm
- Security & DevSecOps:** CI/CD Hardening, Vulnerability Management (Trivy), Google SecOps, Splunk, Penetration Testing, IAM Zero Trust, ESO
- Programming & Automation:** Python, Go, Bash/Shell, JSON Data Pipelines, REST APIs, GitOps (ArgoCD)
- Certifications:** TCM Practical Junior Penetration Tester (PJPT) | ISC2 Certified in Cybersecurity (CC) | AWS Certified Cloud Practitioner

WORK EXPERIENCE

- DevOps Engineer — Seqera Labs** 05/2025 – Present | Toronto, ON
- Architected Terraform and ArgoCD ApplicationSets for declarative multi-environment synchronization, integrating External Secrets Operator (ESO) with AWS Secrets Manager to eliminate manual credential management for 100+ secrets.
 - Engineered keyless CI/CD workflows utilizing GitHub Actions and AWS OIDC federation; integrated automated Trivy and IaC vulnerability assessments into PR gates, mitigating 90% of security risks prior to production release.
 - Diagnosed and resolved cloud networking bottlenecks, including AWS NLB 350-second TCP idle timeouts during multi-gigabyte container image pulls; executed distributed stress testing to ensure build infrastructure scaled under concurrency.
 - Implemented centralized security observability using Google SecOps, Stackdriver, and Cloud Logging, establishing automated threat-detection alerts that slashed Mean Time to Resolution (MTTR) by 75% (from 6 hours to 1.5 hours).
 - Orchestrated high-compute workloads on NVIDIA GPUs and HPC infrastructure, optimizing containerized Python data-processing pipelines to maintain 99.9% availability during resource-intensive computational runs.
 - Conducted internal penetration testing across web applications and platform services, identifying vulnerabilities and implementing remediation workflows prior to deployment.
- Cloud Support Engineer — Seqera Labs** 06/2023 – 05/2025 | Toronto, ON
- Resolved complex L2/L3 infrastructure escalations across AWS, Azure, and GCP, diagnosing compute provisioning bottlenecks and container orchestration failures to resolve 95% of tier-3 issues within a strict 24-hour SLA.
 - Engineered custom Python and Bash automation scripts to parse error logs, validate RBAC/IAM permissions, and automate diagnostic triage, reducing average ticket resolution time (MTTR) by 40%.
 - Hardened enterprise cloud environments across 100+ client accounts by troubleshooting and enforcing zero-trust IAM policies, KMS encryption configurations, and secure VPN network tunnels.
- IT System Administrator — Cloudbeds** 04/2022 – 06/2023 | Toronto, ON
- Administered internal cloud platforms and compute environments, securing network access controls and automating access provisioning to drive a 50% improvement in IT operational efficiency.
 - Architected technical triage procedures and remote service playbooks, streamlining incident escalation pathways to increase first-contact resolution by 20%.
- Infrastructure Administrator — TractionOnDemand** 12/2019 – 03/2022 | Toronto, ON
- Automated IAM workflows in Okta and orchestrated endpoint security incident response (CrowdStrike, Python/Bash), reducing security incidents by 25%. Acted as L3 escalation point for Azure and GCP.

PROJECTS

- Amanah AI Gateway & Pythia CLI — Go, Python, LLM Security** 2026
- Engineered a Go-based runtime security proxy to intercept and validate Large Language Model tool calls, actively preventing prompt-injection attacks; deployed to homelab environment.
 - Developed Pythia, an automated Python CLI tool designed for red-teaming and auditing generative AI pipelines against adversarial prompt vulnerabilities (University Dissertation).

EDUCATION

- MSc Computer Science with Cybersecurity** — University of York Expected 2026
- BSc (Hons) Sport and Exercise Science** — Buckinghamshire New University July 2017